

Statistical Inference and Multivariate Analysis (MA324)

LECTURE SLIDES
Topic 03
Multiple Linear Regression

Jan-May 2026

Multiple Linear Regression

- In general, the response (y) may be related to p regressors (input variables/predictors).
- The model

$$y_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip} + \epsilon_i, i = 1, \dots, n$$

is called **multiple linear regression**. A regression model with p regressors.

- The parameters $\beta_j, j = 0, 1, 2, \dots, p$ are called regression coefficients.
- $\beta_j, j = 0, 1, 2, \dots, p$ represents the change in the average value of the response for a unit change in j^{th} regressor keeping other regressors fixed.

- Data (of sample size n): $(y_i, x_{i1}, x_{i2}, \dots, x_{ip}), i = 1, \dots, n$
- Scalar notation becomes cumbersome – so matrix notation is used
- Let

$$y = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}, X = \begin{pmatrix} 1 & x_{11} & \dots & x_{1p} \\ \vdots & \vdots & & \vdots \\ 1 & x_{n1} & \dots & x_{np} \end{pmatrix}, \beta = \begin{pmatrix} \beta_0 \\ \vdots \\ \beta_p \end{pmatrix}, \epsilon = \begin{pmatrix} \epsilon_1 \\ \vdots \\ \epsilon_n \end{pmatrix}$$

- Then we can write the model in a more compact form:

$$y_{n \times 1} = X_{n \times (p+1)} \beta_{(p+1) \times 1} + \epsilon_{n \times 1}$$

- X is called the ***design matrix***

Error Assumptions

- As before, ϵ_i 's satisfy
 - $E(\epsilon_i) = 0$ for $i = 1, \dots, n$
 - $Var(\epsilon_i) = \sigma^2$ for $i = 1, \dots, n$
 - $Cov(\epsilon_i, \epsilon_j) = 0$ for all $i \neq j$
- Equivalently, the assumptions can be written as
 - $E(\epsilon) = \mathbf{0}$
 - $Var(\epsilon) = \sigma^2 I_n$

Estimation of Regression Coefficients:

- Consider the objective function

$$\begin{aligned}Q(\boldsymbol{\beta}) &= Q(\beta_0, \beta_1, \dots, \beta_p) \\&= \sum_{i=1}^n \left(y_i - \beta_0 - \beta_1 x_{i1} - \dots - \beta_p x_{ip} \right)^2 \\&= (y - X\boldsymbol{\beta})^T (y - X\boldsymbol{\beta}) \\&= (y^T y - 2\boldsymbol{\beta}^T X^T y + \boldsymbol{\beta}^T X^T X \boldsymbol{\beta})\end{aligned}$$

- The LSEs of $\beta_0, \beta_1, \dots, \beta_p$ are $\underset{\boldsymbol{\beta}}{\operatorname{argmin}} Q(\boldsymbol{\beta})$.

Vector Derivative : Linear Function

- Let x be an $n \times 1$ column vector (the variable).
- Let a be a constant $n \times 1$ column vector.
- The derivative of a scalar-valued linear function with respect to the vector x is

$$\frac{\partial}{\partial x}(a^T x) = a.$$

Vector Derivative: Quadratic Function

- Let A be a constant $n \times n$ matrix.
- The derivative of a quadratic form with respect to the vector x is

$$\frac{\partial}{\partial x} (x^T A x) = (A + A^T) x.$$

- If A is a symmetric matrix, then the above formula simplifies to

$$\frac{\partial}{\partial x} (x^T A x) = 2Ax.$$

Estimation of Model Parameters:

- Differentiating $Q(\beta)$ with respect to β and setting the derivative to zero gives the following *normal equations*:

$$X^T X \beta = X^T y.$$

- The above system of linear equations in β is always consistent.
- If the matrix $X^T X$ is *invertible* (i.e. if X is of *full rank*), then the LSE is unique and given by

$$\hat{\beta} = (X^T X)^{-1} X^T y.$$

Fitted Value:

- The fitted value of response corresponding to regression $x = (1, x_1, \dots, x_p)$ is

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \dots + \hat{\beta}_p x_p$$

- Then $\hat{\mathbf{y}} = (\hat{y}_1, \hat{y}_2, \dots, \hat{y}_n)^T = X\hat{\beta} = X(X^T X)^{-1} X^T \mathbf{y} = H\mathbf{y}$, where, $H = X(X^T X)^{-1} X^T$ is called **hat-matrix**.
- The residuals:

$$\mathbf{e} = \begin{bmatrix} e_1 \\ \vdots \\ e_n \end{bmatrix} = \begin{bmatrix} y_1 - \hat{y}_1 \\ \vdots \\ y_n - \hat{y}_n \end{bmatrix} = \mathbf{y} - \hat{\mathbf{y}} = (I - H)\mathbf{y}$$

Properties of LSE of β

- $\hat{\beta}$ is a linear function of y
- $\hat{\beta}$ is unbiased estimator of β . That is, $E(\hat{\beta}) = \beta$
- $Var(\hat{\beta}) = \sigma^2(X^T X)^{-1}$
- $\hat{\beta}$ is the BLUE of β .

Estimation of Error Variance (σ^2)

- It can be shown that

$$E(SS_{Res}) = (n - p - 1)\sigma^2$$

- Hence, $\hat{\sigma}^2 = \frac{SS_{Res}}{n-p-1} = MS_{Res}$ is an unbiased estimator of σ^2 . Here MS_{Res} is residual mean square.
- Observed value of $\hat{\sigma}^2 = \frac{SS_{Res}}{n-p-1}$ is called **Residual variance**. It's **square root is called residual standard error**.
- $\frac{(n-p-1)MS_{Res}}{\sigma^2} \sim \chi^2_{n-p-1}$.
- Here,

$$SS_{Res} = \sum_{i=1}^n e_i^2 = y^T y - \hat{\beta}_1^T X^T y,$$

Hypothesis Testing: Test for Significance of Regression

- Assumption: $\epsilon \sim N_n(0, \sigma^2 I_n)$.
- Want to test the hypothesis if there is a linear relationship between the response y and any of the regressors $x_1, \dots, x_n$:

$$H_0 : \beta_1 = \beta_2 = \dots = \beta_p = 0 \text{ ag. } H_1 : \beta_j \neq 0 \text{ for atleast one } j = 1, \dots, p$$

- The test statistic is

$$F_0 = \frac{SS_{Reg}/p}{\hat{\sigma}^2} = \frac{SS_{Reg}/p}{SS_{Res}/(n-p-1)} \sim F_{p,n-p-1}, \text{ under } H_0.$$

- Reject H_0 iff $F_0 > F_{p,n-p-1;\alpha}$ (at level α)

Hypothesis Testing: Significance of regression coefficients

- Want to test

$$H_0 : \beta_j = 0 \text{ ag. } H_1 : \beta_j \neq 0$$

- Note that

$$\hat{\beta} \sim N_p \left(\beta, \sigma^2 (X^T X)^{-1} \right) \implies \hat{\beta}_j \sim N \left(\beta_j, \sigma^2 C_{jj} \right),$$

where C_{jj} is the j -th diagonal element of $(X^T X)^{-1}$

- The test statistic is

$$t_0 = \frac{\hat{\beta}_j}{\sqrt{MS_{Res} C_{jj}}} \sim t_{n-p-1} \text{ under } H_0$$

- Reject H_0 iff $|t_0| > t_{n-p-1, \alpha/2}$ (at level α)

Test of contribution of a subset of the regressors



$$y = X\beta + \epsilon \implies y = (X_1, X_2) \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix} + \epsilon$$

- Want to test:

$$H_0 : \beta_2 = 0 \text{ ag. } H_1 : \beta_2 \neq 0$$

- Based on the full model ($y = X\beta + \epsilon$), $\hat{\beta} = (X^T X)^{-1} X^T y$ and $SS_{Reg}(\beta)$ has p degrees of freedom.
- To find the contribution of β_2 , fit the model assuming $\beta_2 = 0$.
- The reduced model is $y = X_1\beta_1 + \epsilon$, $\hat{\beta}_1 = (X_1^T X_1)^{-1} X_1^T y$ and $SS_{Reg}(\beta_1)$ has $p - r$ degrees of freedom. Where r denotes the number of components in β_2

Test of contribution of a subset of the regressors

- $SS_{Reg}(\beta_2 \mid \beta_1) = SS_{Reg}(\beta) - SS_{Reg}(\beta_1)$ can be used as a measure of contribution of β_2
- Note that if β_2 has significant contribution then $SS_{Reg}(\beta_2 \mid \beta_1)$ is large
- Therefore, the test statistic is

$$F_0 = \frac{SS_{Reg}(\beta_2 \mid \beta_1)/r}{\hat{\sigma}^2} = \frac{SS_{Reg}(\beta_2 \mid \beta_1)/r}{SS_{Res}/(n-p-1)} \sim F_{r,n-p-1}, \text{ under } H_0$$

- Reject H_0 iff $F_0 > F_{r,n-p-1;\alpha}$ (at level α)

Confidence intervals for β_j :

- Confidence Interval of individual regression coefficient $\hat{\beta}_j$

- Pivot is

$$\frac{\hat{\beta}_j - \beta_j}{\sqrt{MS_{Res}C_{jj}}} \sim t_{n-p-1},$$

where C_{jj} is the diagonal element of $(X^T X)^{-1}$ matrix.

- A $100(1 - \alpha)\%$ CI for β_j is

$$\left[\hat{\beta}_j \pm t_{n-p-1, \alpha/2} \sqrt{MS_{Res}C_{jj}} \right].$$

Confidence interval for mean response:

- Let $x_0 = \begin{pmatrix} 1 \\ x_{01} \\ \vdots \\ x_{0p} \end{pmatrix}$ be a value of the regressor vector
- The mean response at x_0 is $x_0^T \beta$.
- Pivot is $\frac{\hat{y}_0 - x_0^T \beta}{\sqrt{MS_{Res} x_0^T (X^T X)^{-1} x_0}} \sim t_{n-p-1}$.
- A $100(1 - \alpha)\%$ CI for $x_0^T \beta$ is

$$\left[\hat{y}_0 \pm t_{n-p-1, \alpha/2} \sqrt{MS_{Res} x_0^T (X^T X)^{-1} x_0} \right].$$